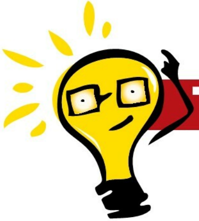

TIPS & TRICKS

Debug your application with the GDB:GNU debugger

The purpose of a debugger such as GDB is to allow us to see what is going on 'inside' a program while it executes, or what the program was doing at the moment it crashed.

We can use GDB to debug programs written in C or C++. GDB is invoked with the shell command *gdb*.

Run the following commands to debug a program:

1. Compile the file using the command:

```
g++ -g xxx.cpp (where xxx is any file)
```

2. Execute the program as follows:

```
gdb ./a.out
```

3. The program will enter into:

```
debugging mode
(gdb)
```

4. Type 'b' (set breakpoint) followed by the function name and press 'Enter'.

```
b main (main function) or b xxx (where xxx is any function name)
```

5. Type 'r' (start the debug program) and press 'Enter', which will display the following:

```
Starting program: /root/a.out
Breakpoint 1, main () at xxx.cpp:7 (7 is line number in the file)
```

6. Type 'n' (step program, proceeding through subroutine calls), till the program exits normally.
7. Other commands that can be used for debugging are:
   s: step program until it reaches a different source line
   bt: backtrace, which displays the program stack
   q: quit debugger

– Sowmya Mitra Attaluri, mitra2270@gmail.com

A command that checks the memory used by each process on the server

We usually check memory utilisation using the *top* command, but it gives the result in a percentage. The following command will give the exact value of the memory used by a process and also sort it according to usage:

```
[root@centos ~]# ps -e -orsz=,vsz=,args= | sort -n | pr
-TW$COLUMNS
```

The following command will give the top 10 commands, according to the non-swapped physical memory that a command has used:

```
[root@centos ~]# ps -e -orsz=,vsz=,args= | sort -n | pr
-TW$COLUMNS | tail -10

10960 54804 /usr/libexec/gdm-user-switch-applet -oaf-
activate-iid=OAFIID:GNOME_FastUserSwitchApplet_Factory -
oaf-ior-fd=28
11164 43852 /usr/libexec/wnck-applet -oaf-activate-
iid=OAFIID:GNOME_Wncklet_Factory -oaf-ior-fd=18
11276 57460 nm-applet -sm-disable
12308 52060 gnome-terminal
12724 45156 gnome-panel
13140 46920 /usr/libexec/clock-applet -oaf-activate-
iid=OAFIID:GNOME_ClockApplet_Factory -oaf-ior-fd=34
15208 33640 python /usr/share/system-config-printer/applet.py
17404 67636 /usr/bin/gnote -panel-applet -oaf-activate-
iid=OAFIID:GnoteApplet_Factory -oaf-ior-fd=21
18320 29620 /usr/bin/Xorg :0 -nr -verbose -auth /var/run/
gdm/auth-for-gdm-oQa9lk/database -nolisten tcp vt1
19304 73780 nautilus
[root@centos ~]#
```

– Shailesh Vaidya, skimeer@gmail.com

Discover the power of the 'history' command

1. How to display the time stamp using HISTTIMEFORMAT

Typically, when you type 'history' from command line, it displays the command# and the command. For auditing